

Erick Torres Aguirre

Physics Engineer



About me

Physics engineering student passionate about software development, scientific computing, and data analysis. I enjoy solving complex problems, building clean and efficient tools, and learning new technologies every day.

personal

Name: Erick
Nationality: Mexican
Born: 2003
Location: Delicias,
Chihuahua, México

Areas of specialization

Web Development
Data Analysis
Scientific Computing
Algorithms
Machine Learning

Interests

Competitive programming
Quantum computing
AI
Videogames

Languages

Spanish (native)
English (B2)

SHORT RESUMÉ

Almost graduate Physics Engineering student with experience building modern web applications and developing analytical tools using Python, JavaScript, and React. Strong background in mathematics and algorithms, national basic sciences champion and ICPC national finalist. Research experience at CIMAV working with scientific datasets and data processing pipelines. Passionate about creating clean, efficient software and learning new technologies.

PROJECTS

• E-commerce Platform — React, TypeScript, Express

- Built a full e-commerce web application with product catalog, filters.
- Implemented shopping cart with add/remove/update logic.
- Integrated MercadoPago API for real online payments.
- Developed product detail pages with reviews and dynamic content.

• Sorting Algorithm Visualizer — React, TypeScript

- Interactive visualizer displaying multiple sorting algorithms step-by-step.
- Showcases algorithm logic with code snippets and animations.
- Added "Orchestra mode" to compare algorithms simultaneously.

• URL Shortener — SQLite, Turso DB, Node.js

- Created a web service for managing user-personalized short URLs.
- Optimized database structure for fast lookup.

EDUCATION & CERTIFICATES

- 2021–2026 Physics Engineering (Graduating in July 2026)
- 2025 Summer Research at CIMAV
 - Worked on scientific data analysis related to nanoparticle characterization.
 - Assisted in organizing datasets, generating visualizations, and analyzing UV-Vis spectra.
 - Collaborated with researchers to automate repetitive analysis tasks.

ACHIEVEMENTS

- 2025 National champion of basic sciences (ANFEI 2025)
- 2025 National finalist of competitive programming (ICPC México)

SKILLS

- **Programming:** Python, JavaScript, TypeScript, C++, LaTeX
- **Web Development:** React, Node.js, Express, REST APIs
- **Databases:** SQLite, Turso DB
- **Tools:** Git, Linux, Neovim, command-line workflows
- **Other:** Algorithms, problem-solving

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